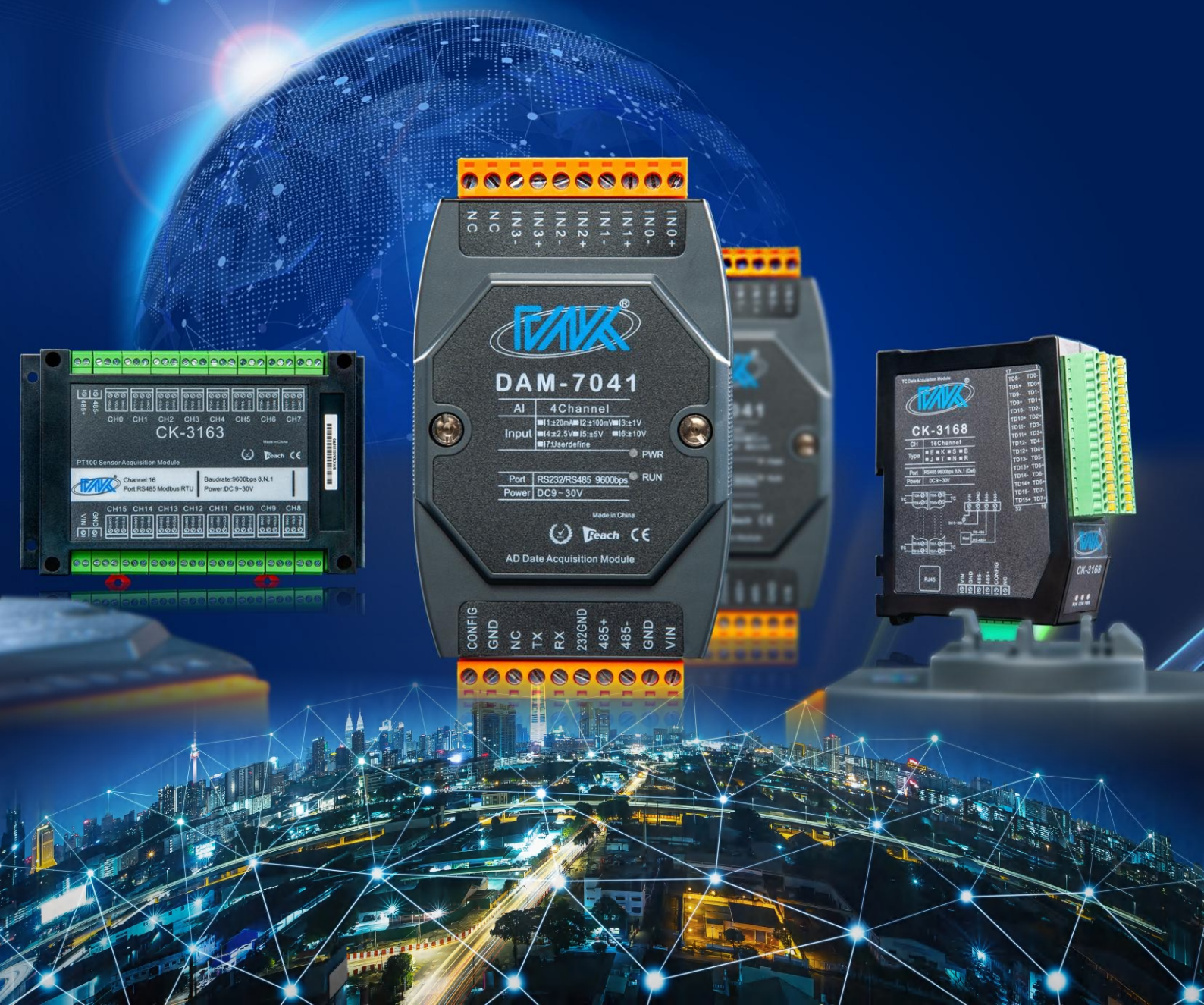
COMPANY PROFILE

Chengkong Electronics,
Professional data collection product supplier。

More than ten years of ingenious quality assurance
and first-class data collection services



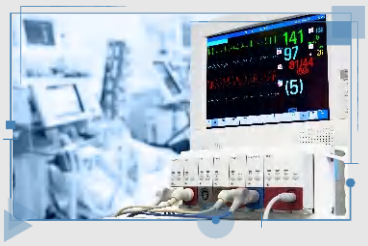
Product Categories

Ⅰ	Analog input module	Ⅱ	Analog output module
Ⅲ	AC input module	Ⅳ	Weighing module
Ⅴ	TC/RTD temperature acquisition module	Ⅵ	Analog input and output module
Ⅶ	Switching/digital module	Ⅷ	Develop custom modules
Ⅸ	signal isolator	Ⅹ	Interface conversion module

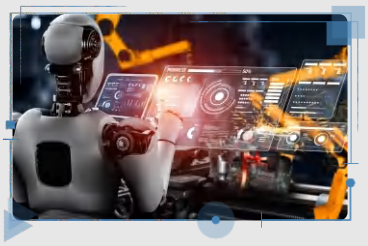
Application areas



Automation equipment



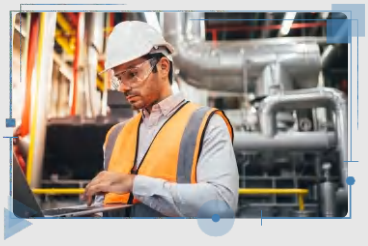
Medical electronics



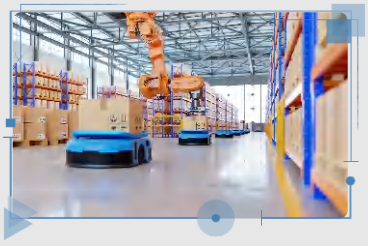
Smart manufacturing



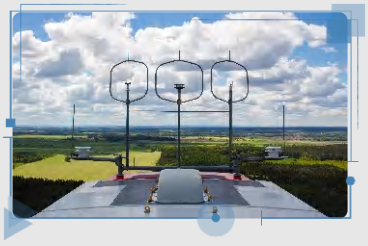
Remote monitoring



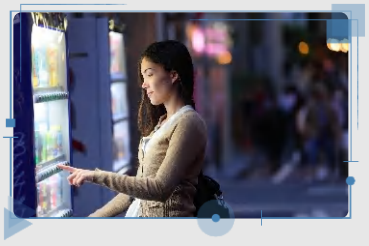
industrial control



Smart warehousing



Instruments anemometer



new retail



8-CH relay output module

Overview

Data Acquisition Module (DAM) is a new generation of modular embedded system based data The Data Acquisition Module (DAM) is a new generation of modular embedded system based data collectors.Easy to install and flexible to use, it can cope with all kinds of on-site applications. The module is equipped with The module is equipped with RS485 interface, which is convenient for communication with PC or PLC.The module is equipped with RS485 interface, which is convenient for communication with PC or PLC.The module is equipped with RS485 interface, which can communicate with PC or PLC individually, and can also be used in a network with multiple 485 modules.

CK-508 is an industrial grade relay output module, which can output 8-channel relay signals; it is suitable for a variety of control systems in industrial sites.

CK-508 adopts magnetic/optical isolation technology, effectively guaranteeing the speed, reliability and safety of data acquisition.speed, reliability and safety of data acquisition.

Technical Parameters

- ◆Embedded real-time operating system
- ◆ Analog output channel: 8 channels
- ◆Contact capacity: 250V 3A
- ◆Wide power supply range： DC 10～30V
- ◆Address/Baud Rate/Check Bit/Stop Bit Configurable
- ◆Support Modbus RTU protocol
- ◆RS485 isolated communication
- ◆ ±15KV ESD protection
- ◆Isolation withstand voltage: DC 2500V
- ◆Operating Temperature Range: -30℃～70℃
- ◆Industrial grade ABS flame retardant housing standard DIN35 rail

Applications

Automation equipment
Remote monitoring and data collection
Intelligent manufacturing/smart factory
Industrial site control
Smart warehousing and monitoring
Medical and industrial control product development
Packaging and material transfer
Electronic product manufacturing

Function configuration

Module Model	CK-508	CK-509
Output Channel	8Relay	8NPN
Input Channel		8OptoCoupler
RS485	Support	Support

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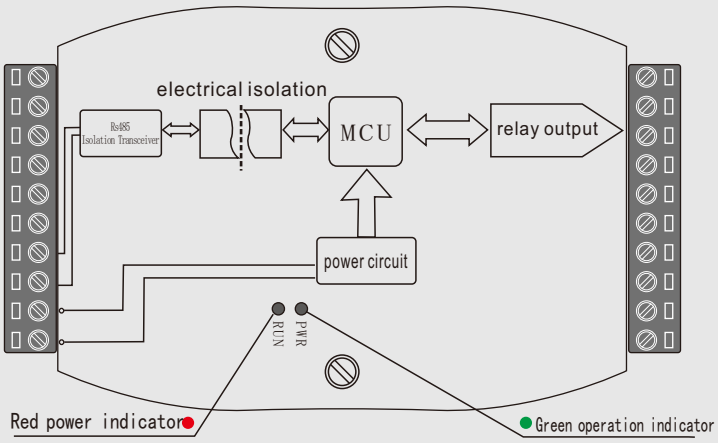
CK-508 8-ch relay output

Contact capacity: 250V 3A
Output: RS485 modbus-RTU



CK-508 is an industrial grade relay output module, which can output 8-channel relaysignals; it is suitable for a variety of control systems in industrial sites.

Module Operating Principle Diagram



Output isolation

All relay output terminals, are isolated from the internal system.

communications isolation

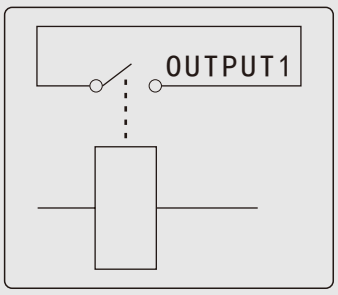
The product adopts an isolated RS485 solution, using DC-DC power supply and magnetic coupling technology to isolate the communication from the system and eliminate common mode interference between communication devices.

surge protection

The module is equipped with a transient suppression circuit, which can effectively suppress various surge pulses and protect the module to work reliably in harsh environments.

Relay output

Output equivalent circuit:



The CK-508 module is equipped with 8 relay output channels; it can be used to directly replace the intermediate relay and is suitable for street light control, water spray control, contactor drive, solenoid valve drive, etc.

Port Information



CK-508Port Description

Port	Port ID	Port Function
1	485+	RS485 signal positive input terminal
2	485-	RS485 signal negative input terminal
3	GND	Power Ground
4	VIN	Power input positive terminal
5	R8	Relay output port 8
6	R8	Relay output port 8
7	R7	Relay output port 7
8	R7	Relay output port 7
9	R6	Relay output port 6
10	R6	Relay output port 6
11	R5	Relay output port 5
12	R5	Relay output port 5
13	R4	Relay output port 4
14	R4	Relay output port 4
15	R3	Relay output port 3
16	R3	Relay output port 3
17	R2	Relay output port 2
18	R2	Relay output port 2
19	R1	Relay output port 1
20	R1	Relay output port 1

Communication interface

CK-508 is equipped with 1 RS485, which can be connected to PLC or other hosts by single module or multiple modules can be connected to PLC or other hosts by networking.

RS485 Connection

CK series modules support standard RS485 interface (see the figure below)

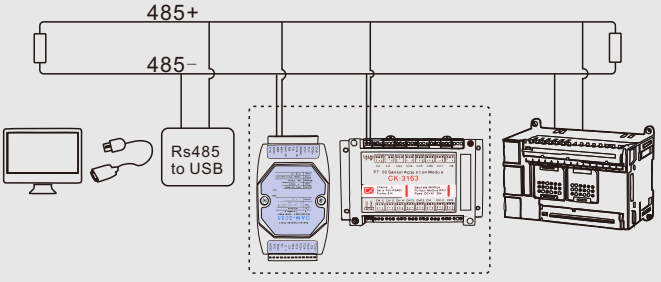
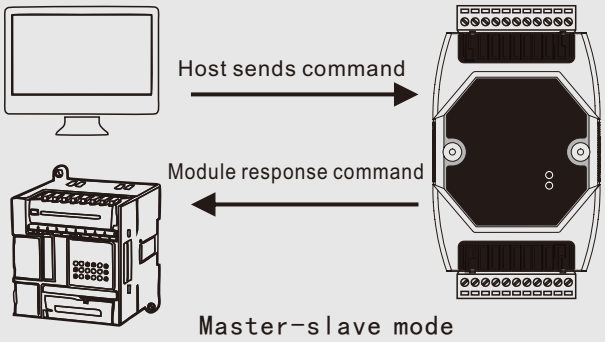


Diagram of CK module connecting to other devices through RS485 interface

Module communication mode

Master-slave mode

The CK-508 module communication mode is usually the master-slave mode (question-answer mode); the host sends commands to the module through the communication interface, and the module responds accordingly after receiving the correct command.



Master-slave mode

Communication parameters (default 9600 8,N,1 address 01)

Mailing address

The communication address range of the CK-508 module is 01H~F7H (1～247), and the module address is factory set to 01; the module communication address can be modified by the user through commands according to site needs. For specific methods, please refer to the corresponding commands.

Communication rate

The CK-508 module RS485 supports baud rates:1200bps, 2400bps, 4800bps, 9600bps, 19200bps, 38400bps, 57600bps, 115200bps; the module communication rate can be modified by the user through commands according to site needs.For specific methods, please refer to the corresponding commands.

Communication Protocol

The CK-508 module supports the industrial standard MODBUS-RTU protocol. After the module is configured as the MODBUS-RTU protocol through MODBUS commands, it can work in the MODBUS slave state. It can communicate with PLCs, RTUs or computers of various brands. The MODBUS commands supported by the CK-508 module are as follows:

Serial number	Order (HEX)	Function	Remark
1	01	Read multiple switch output status (bit)	
2	05	Set the single-channel switch output status (bit)	
3	0F	Set the multi-channel switch output status (bit)	
4	03	Read multiple switch output status (byte)	
5	06	Set the multi-channel switch output status (byte)	
6	10	Set the multi-channel switch output status (byte)	

Modbus protocol is a universal communication protocol that has been widely used in today's industrial control field. Through this protocol, controllers can communicate with each other or with other devices via a network (such as Ethernet).

The MODBUS address of the CK module is assigned as follows:

Order (HEX)	Register address (HEX)	Data Description
01/05/0F	0000	Read and write channel 0 output status
01/05/0F	0001	Read and write channel 1 output status
01/05/0F	0002	Read and write channel 2 output status
01/05/0F	0003	Read and write channel 3 output status
01/05/0F	0004	Read and write channel 4 output status
01/05/0F	0005	Read and write channel 5 output status
01/05/0F	0006	Read and write channel 6 output status
01/05/0F	0007	Read and write channel 7 output status

Modbus commands supported by DO output modules:

Serial number	Order(HEX)	Function	Remark
1	01	Read the set DO output status	
1	05	Write the output status of a single DO output channel	
1	0F	Write the output status of one or more DO output channels	Most commonly used

DO type output module Modbus RTU communication example:

Example	Set DO output status
Module Description	Number of channels: 8, address: 1
Master sends	01 0F 00 00 00 08 01 C3 BE C4
Module Reply	01 0F 00 00 00 08 54 0D
The main station sends analysis	01:Module slave address 0F: Modbus RTUContinuously write coil function code 00 00:0x0000Register start address 00 08:Number of registers written 01:Number of data bytes C3:Output status data, 0xC3 corresponds to binary 0B 11000011. Relay output type: Set state 1 to relay energized. 0C output type: Set state 1 to 0C gate open (connected to GND). BE C4:CRC Check digit
Module reply analysis	01:Module slave address 0F: Modbus RTU Continuously write coil function code 00 00:0x0000 Register start address 00 08:Number of registers written 54 0D:CRC Check digit

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Writing Data	1	1	0	0	0	0	1	1
Channel Number	7	6	5	4	3	2	1	0
Set Status	1	1	0	0	0	0	1	1

Electrical parameters

Unless otherwise specified, the electrical parameters of the CK-508 data acquisition module are the values when Tamb=25℃.

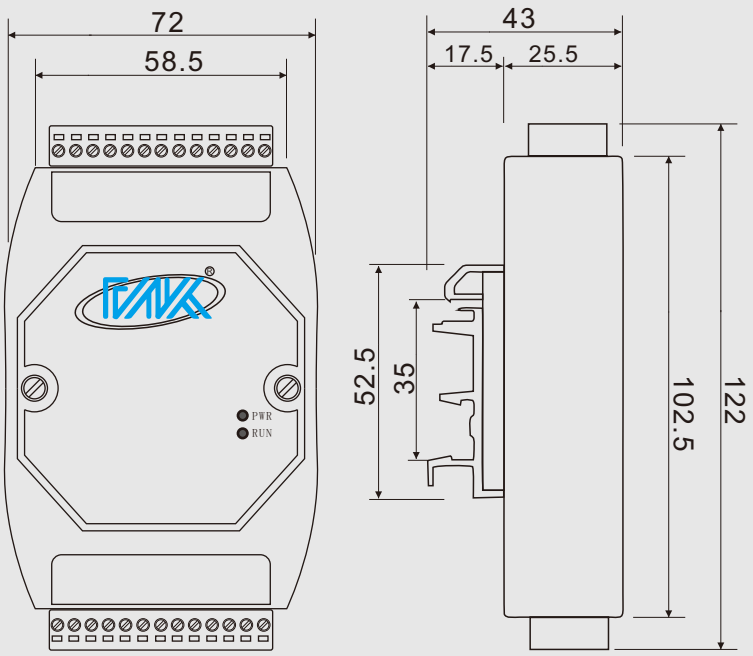
Module parameters

参数	Parameter	最小值 Min	典型值 Typ	最大值 Max	单位 Unit
供电电压	Power Supply	+8	---	+36	V
看门狗复位周期	Watchdog Period		1		S
输入保护	Input Protect		400/32		mA/V

Relay output parameters

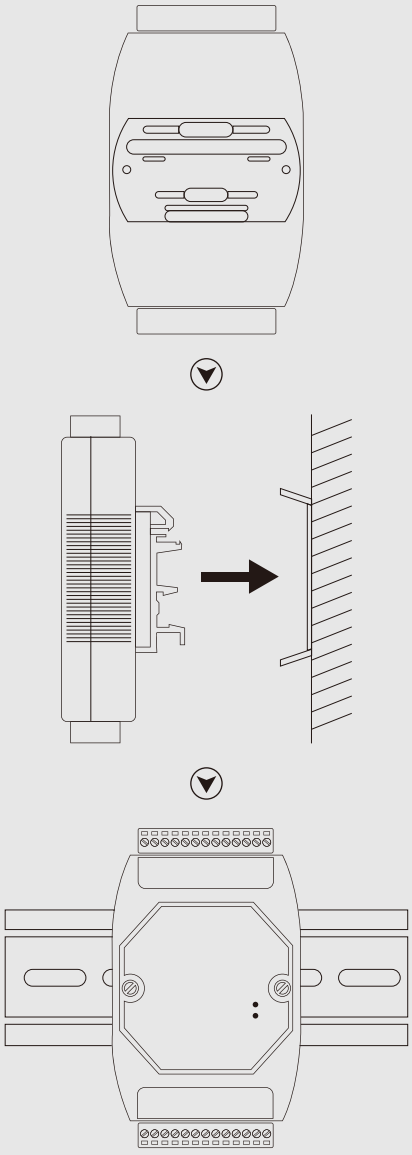
参数	Parameter	最小值 Min	典型值 Typ	最大值 Max
触点容量	Contact Capacity	-	250V 3A	-

Mechanical Dimensions



installation method

CK-508 supports DIN35 rail installation. Users can easily install or remove the module on the rail, providing assistance for industrial site application and installation.



Three guarantees and maintenance instructions

Within five years from the date of sale, if the product is damaged or the product quality is lower than the technical indicators under the conditions of storage, transportation and use, the user can return it to the factory for free repair. If the damage is caused by violation of operating regulations and requirements , the device fee and repair fee shall be paid.

Disclaimer

copyright

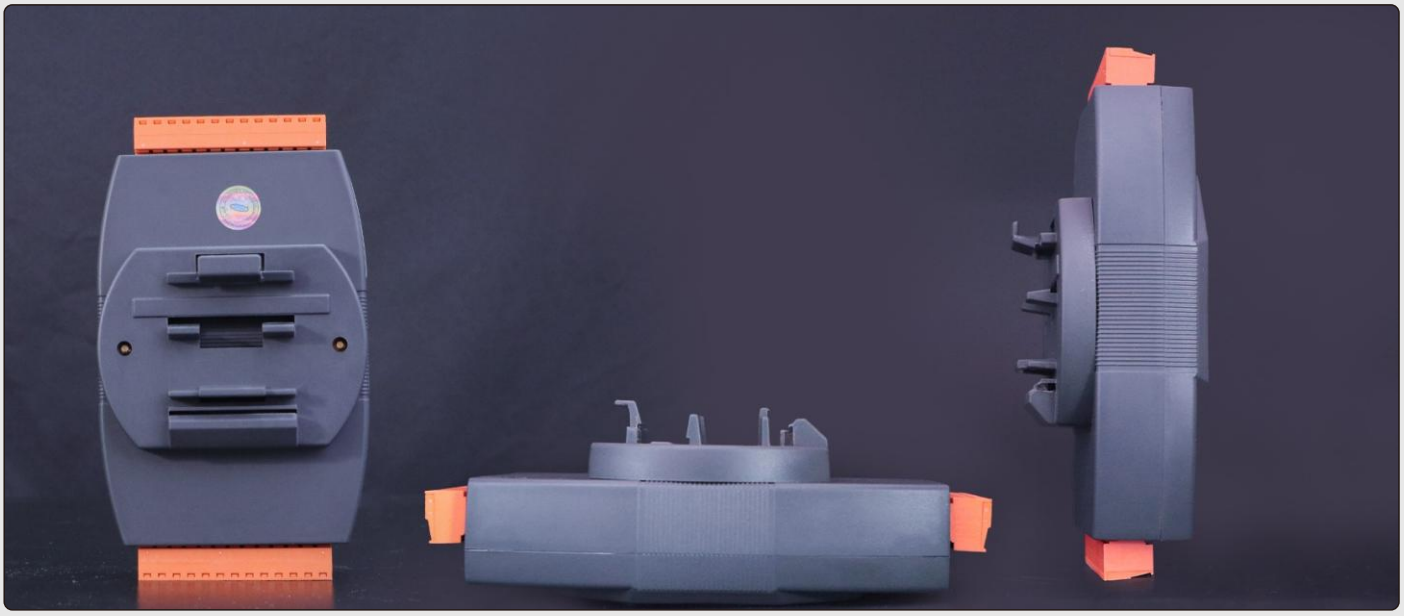
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Product display picture



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Wiring Diagram

